

Yingru Li

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EDUCATION

The Chinese University of Hong Kong (CUHK)

2025

Ph.D. in Computer Science and Information Engineering

Advisor: Zhi-Quan (Tom) Luo.

Committee: Xinyun Chen, Baoxiang Wang, John C.S. Lui, Benjamin Van Roy (Stanford & DeepMind).

Huazhong University of Science and Technology (HUST)

2020

B.E./M.S. in Computer Science. Ranked 1st/134 overall; 1st/26 in Computer Theory and Software.

SELECTED EXPERIENCE

xAI

Singapore

Member of Technical Staff

December 2025 – Present

- Science of RL (Grok 4.2) — I identified and resolved critical training instabilities that benefit every stage of RL training across the entire pipeline, directly enabling the longest stable RL run to date.
- Long-Horizon RL (Grok Next) — I am leading the development of recipes for higher intelligence per token and recursive self-improvement, solving long-horizon credit assignment and enabling continual learning.

ByteDance

Singapore

LLM Algorithm Research Scientist

2025 – December 2025

- I led research on RL training stability and efficiency for LLM reasoning and agents.
- I diagnosed and formalized the RL Collapse phenomenon, tracing instability to implicit off-policiness caused by training-inference mismatch, which induces gradient variance and bias. I developed Trust Region Masking, Dynamic Vocabulary Pruning, the Optimal Token Baseline, and a simple LR scheduling fix — complementary methods that together address these root causes.
- I co-led SimpleTIR for end-to-end multi-turn tool-integrated reasoning and Dr. Kernel for Triton kernel generation via RL.

Tencent AI & Robotics X

Shenzhen, China

Student Research Staff Member, Agent Center

2019 – 2022

- Led the development of HyperAgent (ICML 2024) and HyperDQN (ICLR 2022) for scalable exploration in deep RL; Divergence-Augmented Policy Optimization (NeurIPS 2019) for stable off-policy data reuse.

Microsoft Research Lab – Asia

Beijing, China

Research Intern, Theory Center

2016

- Research on combinatorial optimization and reinforcement/bandit learning algorithms.

SELECTED PUBLICATIONS

Full list at [Google Scholar](#). ★ denotes equal contribution. † denotes corresponding author.

1. The Optimal Token Baseline: Variance Reduction for Long-Horizon LLM-RL

Yingru Li, Jiawei Xu, Ziniu Li, Jiakai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *Under submission to ICML 2026.*

2. Dr. Kernel: Reinforcement Learning Done Right for Triton Kernel Generations

Wei Liu, Jiawei Xu, Yingru Li, Longtao Zheng, Tianjian Li, Qian Liu, Junxian He. *Under submission to ICML 2026.*

3. When Speed Kills Stability: Demystifying RL Collapse from the Training-Inference Mismatch

Jiakai Liu★, Yingru Li★†, Yuqian Fu, Jiawei Wang, Qian Liu, Yu Shen. *Technical Report, 2025.*

4. Logit Dynamics in Softmax Policy Gradient Methods

Yingru Li. *arXiv 2025.*

5. Trust Region Masking for Long-Horizon LLM Reinforcement Learning

Yingru Li, Jiakai Liu, Jiawei Xu, Yuxuan Tong, Ziniu Li, Qian Liu, Baoxiang Wang. *Under submission to ICML 2026.*

6. **Beyond Precision: Training-Inference Mismatch is an Optimization Problem and Simple LR Scheduling Fixes It**
Yaxiang Zhang*, **Yingru Li**^{*,†}, Jiakai Liu, Jiawei Xu, Ziniu Li, Qian Liu, Haoyuan Li. *Under submission to ICML 2026.*
7. **Taming the Tail: Stable LLM Reinforcement Learning via Dynamic Vocabulary Pruning**
Yingru Li, Jiawei Xu, Jiakai Liu, Yuxuan Tong, Ziniu Li, Tianle Cai, Ge Zhang, Qian Liu, Baoxiang Wang. *Under submission to ICML 2026.*
8. **A Note on Hybrid Online RL and Imitation Learning for LLMs: Formulations and Algorithms**
Yingru Li, Ziniu Li, Jiakai Liu. *arXiv 2025.*
9. **Harnessing Uncertainty: Entropy-Modulated Policy Gradients for Long-Horizon LLM Agents**
Jiawei Wang, Jiakai Liu, Yuqian Fu, **Yingru Li**, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, Ke Wang. *Under submission to ICML 2026.*
10. **SimpleTIR: End-to-End Reinforcement Learning for Multi-Turn Tool-Integrated Reasoning**
Zhenghai Xue, Longtao Zheng, Qian Liu, **Yingru Li**, Xiaosen Zheng, Zejun Ma, Bo An. *ICLR 2026.*
11. **Scaling Flaws of Verifier-Guided Search in Mathematical Reasoning**
Fei Yu, **Yingru Li**, Benyou Wang. *arXiv 2025.*
12. **Uncertainty-Aware Search and Value Models: Mitigating Search Scaling Flaws in LLMs**
Fei Yu, **Yingru Li**, Benyou Wang. *arXiv 2025.*
13. **Owl: Optimized Workforce Learning for General Multi-Agent Assistance in Real-World Task Automation**
Mengkan Hu, Yuhang Zhou, Wendong Fan, Yuzhou Nie, Bowei Xia, Tao Sun, Ziyu Ye, Zhaoxuan Jin, **Yingru Li**, Qiguang Chen, Zeyu Zhang, Yifeng Wang, Qianshuo Ye, Bernard Ghanem, Ping Luo, Guohao Li. *NeurIPS 2025.*
14. **Scalable Exploration via Ensemble++**
Yingru Li, Jiawei Xu, Baoxiang Wang, Zhi-Quan Luo. *NeurIPS 2025.*
15. **Q-Star Meets Scalable Posterior Sampling: Bridging Theory and Practice via HyperAgent**
Yingru Li, Jiawei Xu, Lei Han, Zhi-Quan Luo. *ICML 2024.*
16. **Probability Tools for Sequential Random Projection**
Yingru Li. *arXiv 2024.*
17. **Simple, unified analysis of Johnson-Lindenstrauss with applications**
Yingru Li. *arXiv 2024.*
18. **Prior-Dependent Analysis of Posterior Sampling RL with Function Approximation**
Yingru Li, Zhi-Quan Luo. *AISTATS 2024.*
19. **HyperDQN: A Randomized Exploration Method for Deep Reinforcement Learning**
Ziniu Li, **Yingru Li**[†], Yushun Zhang, Tong Zhang, Zhi-Quan Luo. *ICLR 2022.*
20. **Divergence-Augmented Policy Optimization**
Qing Wang*, **Yingru Li***, Jiechao Xiong, Tong Zhang. *NeurIPS 2019.*